

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Replaced polling with SSE for live metrics streaming, building real-time data ingestion into the platform and cutting redundant API calls 90%.
- Engineered a pluggable distributed executor routing pipeline nodes across Docker and AWS Lambda with an S3 artifact store, eliminating redundant recomputation under scale constraints.
- Owned backend services and RESTful APIs (Python/FastAPI) handling millions of daily transactions at 99%+ uptime.
- Built a deployment-gated evaluation framework enforcing quality and regression checks across production outputs, blocking 15% of failed releases before traffic cutover.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Improved retrieval recall 61% to 89% on a 2K-query eval set by rebuilding the chunking pipeline with hybrid dense-sparse retrieval and re-ranking over a vector store (Pinecone).
- Trained and evaluated computer vision models (PyTorch) for production image understanding, running model training and fine-tuning on multi-GPU clusters.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated 25+ production microservices in Java, C++, and SQL-backed distributed systems for 1M+ users, sustaining 99.9% uptime over 3 years on AWS (EC2, RDS, SQS).
- Cut nightly financial data aggregation blocking downstream settlement from 8 minutes to 90 seconds by redesigning SQL with indexed views and batch partitioning.

PROJECTS

LangFetch

tinyurl.com/LangFetch

- Agent decomposing natural-language questions into dependency-ordered SQL subqueries with Pinecone vector retrieval and hybrid re-ranking; deployed as an MCP-compatible API server for enterprise data queries.

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus distributed storage engine handling concurrent writes with linearizable reads and log compaction; companion VPC control plane simulator modeling subnet routing.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Production evaluation and data-generation platform with rubric-based scoring, drift detection, and regression alerting across model versions; FastAPI backend, PostgreSQL.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, Java, C++, Go, SQL

Search & Retrieval: RAG, Vector Databases (Pinecone), Hybrid Dense-Sparse Retrieval, Re-Ranking, Query Decomposition

ML: PyTorch, Model Training/Fine-Tuning, Model Evaluation, Data Generation

Systems: RESTful APIs, Real-Time Ingestion (SSE), Distributed Systems, Linux, AWS, Docker, CI/CD

CERTIFICATIONS

AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer